EAST WEST UNIVERSITY

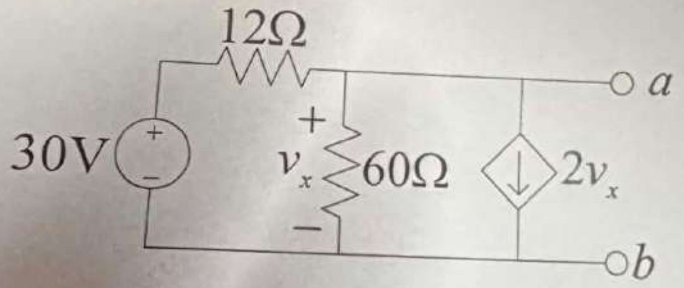
26 April, 2026

Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Final Examination, Spring 2026 Semester

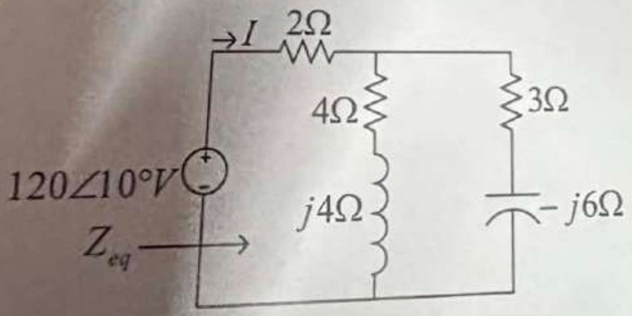
Course: CSE 209 Electrical Circuits, Section-8
Instructor: Musharrat Khan, Senior Lecturer, CSE Department
Full Marks: 30
Time: 1 Hour and 20 Minutes

Note: There are FIVE questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

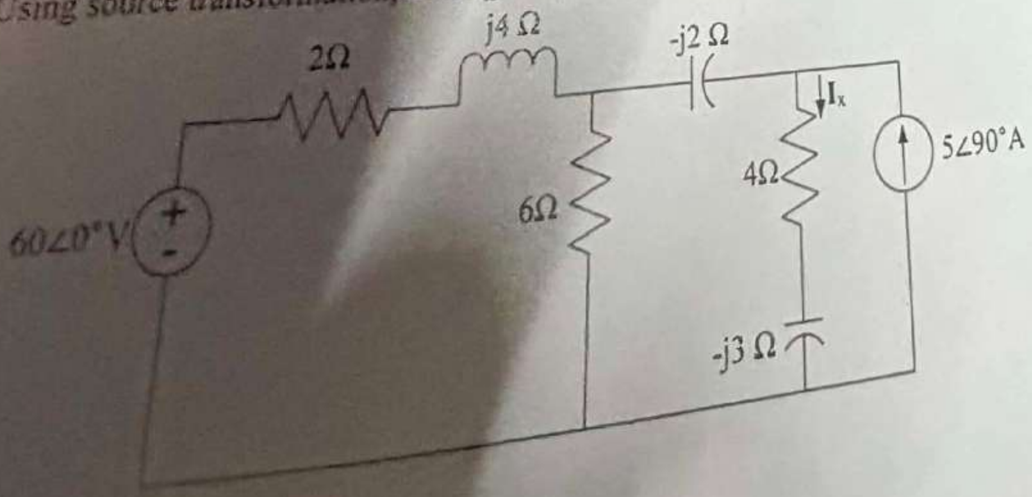
1. **Determine** the Norton's equivalent of the following circuit with respect to terminals a and b. Also **calculate** the maximum power. [CO2,C3, Mark: 6+1=7]



2. **Determine** Z_{eq} and I for the following circuit. [CO1,C2, Mark: 5]



3. Using source transformation, find I_x in the following circuit. [CO3,C3, Mark:6]



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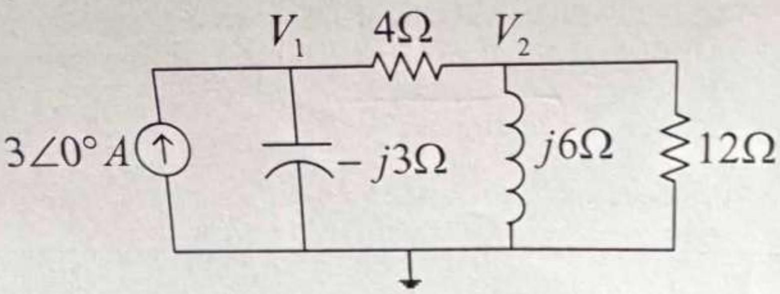
4. Using nodal analysis, **compute V_1** for the following circuit. [CO3,C3, Mark:6]



5. $v(t) = 200\cos(\omega t - 10°)$, $i(t) = 8\cos(\omega t + 30°)$. **Determine** the complex power, the apparent power, the real power, the reactive power, the power factor, and the load impedance. [CO3,C3, Mark:6]